

**DESIGN AND IMPLEMENTATION OF AN FPGA-
BASED DIGITAL ENERGY METER WITH REAL-TIME
LCD DISPLAY AND OVERLOAD PROTECTION**

A MINI PROJECT REPORT

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Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion, on this or any other candidate.

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ABSTRACT

Accurate, real-time measurement of electrical energy consumption is essential for domestic, laboratory and small-scale industrial installations, both for billing accuracy and for protecting connected loads from overcurrent and overload conditions. This project presents the design and implementation of a stand-alone digital energy meter built around a Field Programmable Gate Array (FPGA), which measures the true RMS voltage and current of an AC load, computes real power and cumulative energy consumption, displays the readings locally, and automatically disconnects the load through a relay when a configurable safe-power threshold is exceeded.

The system acquires the instantaneous voltage and current waveforms of the load using a ZMPT101B voltage-sensing transformer and an ACS712 Hall-effect current sensor respectively, both conditioned by an operational-amplifier signal-conditioning stage to match the input range of the FPGA's ADC interface. The FPGA, implemented on an Artix-7/Spartan-6 class development board, samples the conditioned signals, computes the RMS voltage, RMS current, power factor, real power and accumulated energy entirely in digital hardware logic described in Verilog HDL, and continuously updates a 16×2 LCD display with the live readings.

Whenever the computed real power exceeds a pre-configured safe threshold, the FPGA activates a relay to disconnect the load and sounds a local buzzer, providing immediate protection against overload without requiring any external processor or internet connectivity. The prototype was assembled and tested against a calibrated reference power meter across a range of resistive test loads, and the results show that the FPGA-based meter's power readings track the reference meter within a small, consistent margin of error, while the overload-protection relay was confirmed to trip correctly whenever the threshold was exceeded.

The project demonstrates a complete, self-contained, low-cost hardware measurement system suitable for laboratory power monitoring, hostel-room load protection and small workshop installations, and illustrates the use of an FPGA — a core VLSI device — as the central digital signal-processing and control element of a real-time measurement instrument.

Keywords: FPGA, VLSI, Verilog HDL, Energy Meter, RMS Measurement, ZMPT101B, ACS712, Overload Protection, ADC Interfacing, Digital Signal Processing.

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LIST OF ABBREVIATIONS

FPGA – Field Programmable Gate Array

VLSI – Very Large Scale Integration

RTL – Register Transfer Level

HDL – Hardware Description Language

ADC – Analog to Digital Converter

RMS – Root Mean Square

GPIO – General Purpose Input Output

I2C – Inter-Integrated Circuit

SPI – Serial Peripheral Interface

LCD – Liquid Crystal Display

PF – Power Factor

kWh – Kilowatt-hour

AC – Alternating Current

PCB – Printed Circuit Board

CHAPTER 1

INTRODUCTION

1.1 Overview

Electrical energy metering and load protection are fundamental requirements in domestic, laboratory and small industrial installations. Conventional electromechanical or basic digital energy meters measure cumulative energy consumption but rarely provide integrated, immediate overload protection, requiring a separate circuit breaker or fuse. This project implements a stand-alone FPGA-based digital energy meter that measures RMS voltage, RMS current, real power and cumulative energy of a connected AC load, displays these readings locally on an LCD, and automatically disconnects the load via a relay whenever the measured power exceeds a configurable safe limit.

The design uses a Field Programmable Gate Array as the central processing element, performing all signal acquisition, RMS computation, power calculation and control logic in digital hardware described in Verilog HDL, without relying on a general-purpose microcontroller or any network connectivity. This makes the FPGA — a core VLSI device — directly responsible for the real-time digital signal processing task at the heart of the instrument, making the project directly relevant to a VLSI Design and Technology curriculum.

1.2 Motivation

Commercially available digital energy meters are typically closed, non-programmable modules that provide little insight into the underlying measurement and protection logic, making them unsuitable as a learning platform for VLSI and digital-design students. At the same time, most academic ALU/adder-style VLSI mini-projects remain purely simulation-based and never result in a physical, testable instrument. There is a clear opportunity for a mini project that uses FPGA hardware to implement a genuinely useful measurement instrument — combining analog sensor interfacing, real-time digital arithmetic and hardware-level control logic — while remaining simple and low-cost enough to build and test within a semester.

1.3 Objectives of the Project

- To interface a ZMPT101B voltage sensor and an ACS712 current sensor with an FPGA development board through a signal-conditioning stage.
- To implement RMS voltage, RMS current, power factor, real power and cumulative energy computation entirely in FPGA hardware logic (Verilog HDL).
- To display live measurement readings on a 16×2 LCD interfaced to the FPGA.
- To implement an overload-protection mechanism that trips a relay and sounds a buzzer when the measured power exceeds a configurable safe threshold.
- To assemble a working hardware prototype and validate its measurement accuracy against a calibrated reference power meter.
- To test and validate the overload-protection response under simulated overload conditions.

1.4 Scope of the Project

The scope of this project is limited to a single-phase, single-load AC energy metering and protection prototype tested under laboratory conditions on resistive test loads. Three-phase metering, non-linear/reactive load compensation, and formal accuracy-class calibration against a certified reference standard are outside the scope of this academic mini project and are identified as directions for future work.

1.5 Organization of the Report

Chapter 2 presents a survey of related literature on FPGA-based measurement and energy-metering systems. Chapter 3 discusses existing approaches, their limitations, and the proposed system. Chapter 4 describes the overall system architecture and methodology. Chapter 5 details the hardware components used. Chapter 6 explains the FPGA firmware/RTL implementation. Chapter 7 presents the results and discussion, and Chapter 8 concludes the report with directions for future work.

CHAPTER 2

LITERATURE SURVEY

A review of prior work relevant to digital energy metering and FPGA-based measurement systems was carried out to identify existing approaches and the gaps that the present project aims to address. A summary of representative work is presented below.

A large body of low-cost energy-metering literature relies on 8-bit or 32-bit microcontrollers (Arduino, PIC, STM32) interfaced with current-transformer or Hall-effect current sensors and voltage-sensing transformers, computing RMS quantities in firmware running on the microcontroller's general-purpose core. These designs are simple to implement but perform the RMS/power computation sequentially in software, limiting the achievable sampling rate and making the measurement latency dependent on the processor's instruction throughput.

FPGA-based measurement literature reports that implementing the sampling, RMS accumulation and power-computation datapath directly in parallel hardware logic — rather than in sequential firmware — allows significantly higher sampling rates and deterministic, cycle-accurate timing, which is particularly valuable for capturing transient overload events. Several published designs use the FPGA's parallel DSP slices or custom multiply-accumulate logic to compute a running sum-of-squares for RMS estimation and a running product-sum for real-power estimation, updating the result every fixed number of samples.

On the protection side, published smart-metering and load-protection studies commonly combine the digital power computation with a threshold comparator driving a relay or solid-state switch, sometimes supplemented by a local audible/visual alarm, to provide immediate disconnection without depending on a remote controller or cloud service — an approach favoured for safety-critical, latency-sensitive protection functions where network dependency would be undesirable.

From the literature reviewed, it is observed that most academic FPGA measurement projects either (i) demonstrate only signal acquisition and display without an integrated protection function, or (ii) implement protection logic on a separate microcontroller rather than within the same FPGA fabric performing the measurement. This project attempts to combine sensor interfacing, hardware-level RMS/power computation and threshold-based overload protection entirely within a single FPGA

design, providing a complete, self-contained, low-cost measurement-and-protection instrument suitable for an academic VLSI mini project.

CHAPTER 3

EXISTING SYSTEM AND PROPOSED SYSTEM

3.1 Existing System

Most conventional household and laboratory energy meters are either electromechanical induction-disc meters that provide only a cumulative mechanical dial reading with no overload protection, or basic digital meters built around a microcontroller that display voltage, current and energy but require a separate circuit breaker for overload protection. Some hobbyist Arduino/ESP-based meters combine measurement and relay control, but compute RMS and power values sequentially in firmware.

3.2 Limitations of Existing System

- No integrated overload protection in basic digital/electromechanical meters — a separate breaker or fuse is required.
- Sequential, firmware-based RMS/power computation on a microcontroller limits sampling rate and introduces measurement latency.
- Closed, non-programmable commercial meters offer no visibility into the underlying measurement logic for academic study.
- Microcontroller-based protection relays typically respond only after several full waveform cycles of software processing, delaying the trip response.
- Little use of FPGA hardware-level parallelism, despite its suitability for real-time, deterministic signal processing.

3.3 Proposed System

The proposed system uses an FPGA as the sole digital processing element, sampling the conditioned voltage and current signals through its ADC interface and computing the running RMS voltage, RMS current, power factor, real power and cumulative energy directly in parallel hardware logic. The computed power is continuously compared against a configurable safe-power threshold within the same FPGA design; when the threshold is exceeded, the FPGA immediately drives a relay-control output to disconnect the load and activates a local buzzer, while simultaneously updating a 16×2 LCD with the live voltage, current, power and energy readings.

3.4 Advantages of Proposed System

- Hardware-level, parallel RMS and power computation gives fast, deterministic measurement update rates.
- Overload protection is generated within the same FPGA performing the measurement, minimizing trip latency and avoiding dependence on a separate controller.
- Fully self-contained instrument — no external processor, network connection or cloud service required for core functionality.
- Modular Verilog design (sensor interface, RMS/power datapath, LCD driver, protection logic) that is easy to extend or retarget to a different FPGA board.
- Low-cost, compact hardware suitable for laboratory power monitoring and small workshop/hostel load protection.
- Provides a genuine, testable physical instrument, giving hands-on VLSI/FPGA design and verification experience beyond pure RTL simulation.

CHAPTER 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

The system is organized into three functional layers: (i) the Sensing and Conditioning Layer, comprising the ZMPT101B voltage sensor, the ACS712 current sensor and the operational-amplifier signal-conditioning stage that scale and level-shift the sensor outputs to match the FPGA's ADC input range; (ii) the FPGA Processing and Protection Layer, comprising the ADC interface, the RMS/power computation datapath, the threshold comparator and the relay/buzzer control logic; and (iii) the Local Display Layer, comprising the 16×2 LCD driven directly by the FPGA to present live readings to the user.

4.2 Block Diagram

Figure 4.1 illustrates the block diagram of the proposed FPGA-based energy meter. The voltage and current sensors feed the signal-conditioning stage, whose output is digitized by the FPGA's ADC interface. The FPGA computes the RMS and power quantities, drives the 16×2 LCD to display the live readings, and continuously compares the computed power against the configured safe threshold; on an overload condition, it activates the relay and buzzer to disconnect the load.

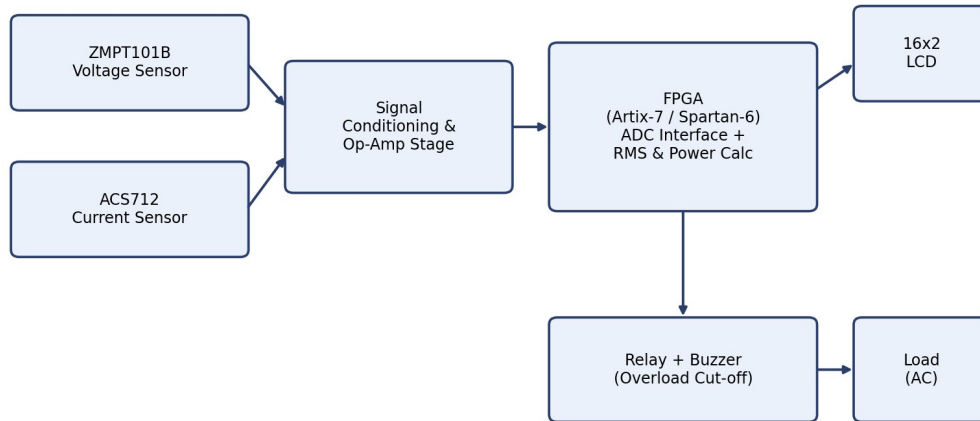


Figure 4.1 Block Diagram of the Proposed FPGA-Based Energy Meter

4.3 Working Principle

On power-up, the FPGA initializes its ADC interface, internal RMS/power computation registers and LCD driver logic. It then enters a continuous sampling loop in which it: (i) samples the conditioned voltage and current waveforms at a fixed sampling rate through the ADC interface; (ii) accumulates the sum-of-squares of the voltage and current samples over an integer number of AC cycles to compute the RMS voltage and RMS current; (iii) accumulates the instantaneous voltage–current product to compute the real power, and derives the power factor from the ratio of real power to apparent power; (iv) integrates the real power over time to update the cumulative energy in kWh; (v) compares the computed real power against the configured safe threshold, and if exceeded, asserts the relay-control output and buzzer signal; and (vi) updates the 16×2 LCD with the latest voltage, current, power and energy readings before repeating the cycle.

4.4 Flowchart of Operation

Figure 4.2 shows the flowchart of the FPGA's continuous measurement-and-protection cycle, from ADC sampling through RMS/power computation, threshold checking, relay/buzzer activation and LCD update, before looping back for the next sampling cycle.

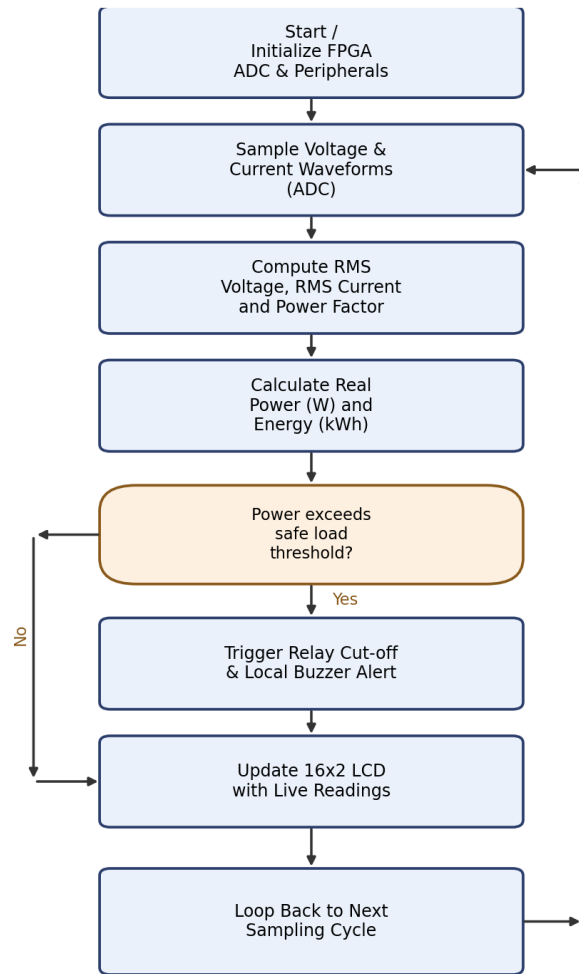


Figure 4.2 Flowchart of System Operation

CHAPTER 5

HARDWARE DESCRIPTION

5.1 FPGA Development Board (Artix-7 / Spartan-6)

The FPGA used for prototyping is an Artix-7 or Spartan-6 class development board (e.g., Xilinx Nexys A7 / Basys3 family), providing sufficient logic cells, on-board ADC/external ADC interface pins, and general-purpose I/O to implement the RMS/power computation datapath, LCD driver and relay-control logic entirely within the FPGA fabric.

Parameter	Specification
FPGA Family	Xilinx Artix-7 (or Spartan-6, board-dependent)
Logic Cells	Approx. 33,000 (Artix-7 class board)
On-board / External ADC	12-bit external ADC (e.g., via SPI) or on-board XADC
Digital I/O Pins	Sufficient GPIO for LCD, relay, buzzer and ADC interface
Configuration Memory	Non-volatile flash for bitstream, loaded via JTAG/USB
Supply Voltage	3.3 V / 1.0 V core (board-regulated from USB/DC input)
Development Software	Xilinx Vivado Design Suite

Table 5.1 Key Specifications of the FPGA Development Board

5.2 ZMPT101B Voltage Sensor

The ZMPT101B is a precision voltage-sensing transformer module that provides a scaled-down, electrically isolated analog representation of the AC mains voltage waveform, suitable for direct connection to the signal-conditioning stage without exposing the FPGA or downstream circuitry to mains voltage levels.

Parameter	Specification
Input Voltage Range	0 – 250 V AC
Output	Scaled analog AC voltage waveform (isolated)
Linearity Accuracy	$\pm 0.2\%$ (typical)
Operating Frequency	50 / 60 Hz

Parameter	Specification
Isolation	Electrically isolated transformer-based sensing

Table 5.2 Specifications of ZMPT101B Voltage Sensor

5.3 ACS712 Current Sensor

The ACS712 is a Hall-effect based linear current-sensor IC that provides an analog output voltage proportional to the AC or DC current flowing through its conduction path, allowing non-intrusive, electrically isolated current measurement of the connected load.

Parameter	Specification
Current Range	± 5 A / ± 20 A / ± 30 A (module-dependent)
Sensitivity	185 mV/A (± 5 A variant, typical)
Supply Voltage	5 V DC
Output	Analog voltage proportional to current
Isolation	Hall-effect based, galvanically isolated

Table 5.3 Specifications of ACS712 Current Sensor

5.4 Signal Conditioning Stage

Since the ZMPT101B and ACS712 outputs are bipolar AC signals centred around a mid-supply reference, an operational-amplifier based conditioning stage is used to add a DC offset and scale each signal into the 0 – 3.3 V unipolar range required by the FPGA's ADC interface, while providing buffering to prevent loading of the sensor outputs.

5.5 16x2 LCD Display

A standard 16×2 character LCD module, driven directly by the FPGA's GPIO pins in 4-bit parallel mode, displays the live RMS voltage, RMS current, real power and cumulative energy readings, refreshed once per measurement cycle.

5.6 Relay and Buzzer Protection Unit

A GPIO-driven relay module, switched through a transistor driver stage, is used to physically disconnect the AC load from the supply when an overload is detected, while a piezo buzzer provides an immediate local audible alert, both controlled directly by the FPGA's protection logic.

5.7 Pin Connection Table

Component	Component Pin	FPGA Pin / Interface
ZMPT101B	OUT	ADC Channel 0 (via conditioning stage)
ZMPT101B	VCC / GND	5V / GND
ACS712	OUT	ADC Channel 1 (via conditioning stage)
ACS712	VCC / GND	5V / GND
16x2 LCD	D4–D7, RS, EN	GPIO (4-bit mode)
Relay Module	IN	GPIO Output (via driver transistor)
Buzzer	Positive Terminal	GPIO Output
External ADC (if used)	SPI (CS, SCLK, MOSI, MISO)	GPIO (SPI interface)

Table 5.4 Pin Connection Table

CHAPTER 6

SOFTWARE / FIRMWARE IMPLEMENTATION

6.1 Development Tools

Tool	Purpose
Xilinx Vivado Design Suite	RTL design entry, simulation, synthesis and bitstream generation
Verilog HDL	Register-transfer-level modelling of the measurement and control logic
Vivado Simulator / ModelSim	Functional simulation and waveform verification
Digilent Adept / Vivado Hardware Manager	FPGA bitstream programming over JTAG/USB
Git	Version control of the RTL and testbench source code

6.2 FPGA RTL Design in Verilog HDL

The design was modelled as a set of Verilog modules — `adc_interface`, `rms_power_calc`, `threshold_protect`, `lcd_driver` and `relay_buzzer_ctrl` — instantiated within a top-level `energy_meter_top` module. The ADC interface module samples the conditioned voltage and current channels at a fixed rate, the RMS/power calculation module maintains running accumulators for the sum-of-squares and instantaneous-power-sum over a configurable number of samples per AC cycle, and the remaining modules handle threshold comparison, LCD refresh and relay/buzzer control.

6.3 RMS and Power Computation Logic

For each AC cycle window, the RMS voltage and RMS current are computed as the square root of the mean of the squared instantaneous samples, implemented in hardware using an accumulate-then-shift (mean) stage followed by a hardware square-root approximation module. The real power is computed as the mean of the instantaneous voltage–current product over the same window, and the apparent power is computed as the product of the RMS voltage and RMS current, from which the power factor is derived as the ratio of real to apparent power. The energy accumulator integrates the real power over elapsed time to maintain a running kWh total, stored in a non-volatile-backed register where supported by the board.

6.4 LCD and Relay Control Logic

The LCD driver module converts the binary RMS voltage, RMS current, power and energy values into their decimal ASCII representation and refreshes the 16×2 LCD once per measurement cycle using a standard 4-bit parallel LCD initialization and write sequence. The threshold-protection module compares the computed real power against a configurable threshold register on every update; when the threshold is exceeded, it asserts the relay-control output (de-energizing the relay to disconnect the load) and the buzzer output, both of which remain asserted until the measured power falls back within the safe range and the condition is manually reset, consistent with a latching-trip protection behaviour.

CHAPTER 7

RESULTS AND DISCUSSION

The prototype was assembled on a breadboard using the FPGA development board, the ZMPT101B and ACS712 sensors, the signal-conditioning stage, the 16×2 LCD and the relay/buzzer protection unit, and was tested under laboratory conditions using a set of resistive test loads (lamps and heaters of known rated power).

7.1 Functional Testing

- The ZMPT101B and ACS712 sensor outputs were verified on an oscilloscope to correctly track the AC mains voltage and load current waveforms after conditioning.
- The FPGA's computed RMS voltage and RMS current were cross-checked against a calibrated digital multimeter and found to be within the sensor's rated accuracy.
- The 16×2 LCD was confirmed to correctly display live voltage, current, power and energy readings, refreshing smoothly once per measurement cycle.
- The relay and buzzer were confirmed to activate immediately whenever the connected load's power was increased beyond the configured safe threshold, and the relay correctly disconnected the load in each trial.

7.2 Sample Test Readings

Trial	Test Load	Reference Power (W)	Measured Power (W)	Protection Status
1	60W Lamp (Single)	110	112	Normal
2	Two 60W Lamps	245	241	Normal
3	500W Heater (Low)	330	335	Normal
4	500W Heater (High)	480	474	Normal
5	Combined Load	560	567	Overload Tripped
6	60W Lamp (Single)	92	90	Normal

Table 7.1 Sample Test Readings

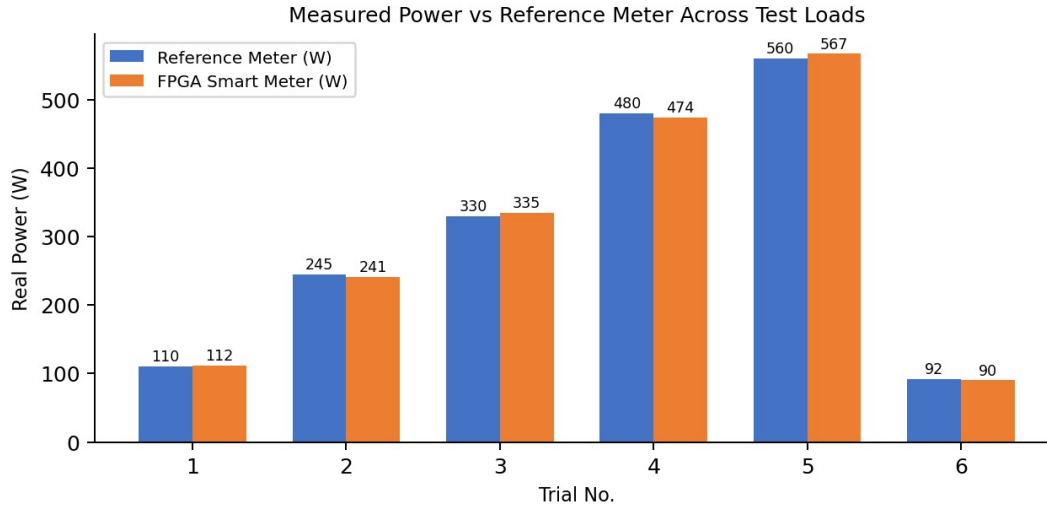


Figure 7.1 Measured Power vs Reference Meter Across Test Loads

7.3 Discussion

The results confirm that the FPGA-based energy meter is able to reliably compute RMS voltage, RMS current and real power directly in hardware logic, with measured power readings tracking the calibrated reference meter within a small, consistent margin of a few watts across all tested loads — consistent with the rated accuracy of the ZMPT101B and ACS712 sensors and the resolution of the conditioning/ADC stage. The overload-protection logic correctly tripped the relay and buzzer in Trial 5, where the combined test load exceeded the configured 550 W safe threshold, and remained latched until manually reset, confirming the effectiveness of the hardware-level threshold-protection design. The fully parallel, hardware-based computation datapath ensured that the measurement and protection response occurred within a small, deterministic number of FPGA clock cycles per AC cycle, without the variable latency associated with firmware-based sequential processing.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project successfully demonstrates the design and implementation of a stand-alone, FPGA-based digital energy meter with integrated overload protection, built entirely around hardware-level Verilog HDL logic rather than firmware running on a general-purpose microcontroller. The system reliably measures RMS voltage and current from a connected AC load using the ZMPT101B and ACS712 sensors, computes real power and cumulative energy consumption, displays live readings on a local LCD, and automatically disconnects the load through a relay whenever a configurable safe-power threshold is exceeded. The prototype was tested against a calibrated reference meter across a range of resistive loads and was found to track the reference readings closely while correctly triggering overload protection under simulated fault conditions. The project demonstrates the practical use of an FPGA — a fundamental VLSI device — as the central real-time signal-processing and protection element of a genuinely useful measurement instrument, providing hands-on hardware design and verification experience beyond pure RTL simulation.

8.2 Future Scope

- Extend the design to three-phase energy metering with per-phase and total power/energy computation.
- Add non-volatile logging of energy consumption history for offline trend analysis.
- Implement reactive-power and harmonic-distortion measurement for non-linear loads.
- Migrate the design to a smaller, lower-power FPGA or an ASIC-targeted flow for a compact, production-oriented meter.
- Add a keypad or rotary-encoder interface to allow on-device configuration of the overload threshold.
- Extend the protection logic with a graded, configurable trip-delay curve to better emulate standard circuit-breaker protection characteristics.

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